

STREPTOCOCCUS

S. pyogenes, Other Haemolytic Streptococci, Enterococcus & Viridans Group

Microbiology (Bacteriology) • Ananthanarayan & Paniker, Chapter 22

1. INTRODUCTION & CLASSIFICATION

- Streptococci are **Gram-positive cocci in chains or pairs**; part of normal flora of humans and animals. Chains form because cocci divide in **one plane** and daughter cells fail to separate (*S. salivarius* forms the longest chains).
- S. pyogenes* causes pyogenic infections with a **tendency to spread** (unlike localised staphylococcal lesions) and **non-suppurative sequelae** — acute rheumatic fever and glomerulonephritis.
- First seen by **Billroth (1874)** in erysipelas (*streptos* = twisted); **Ogston (1881)** isolated them from abscesses; **Rosenbach (1884)** named *S. pyogenes*.

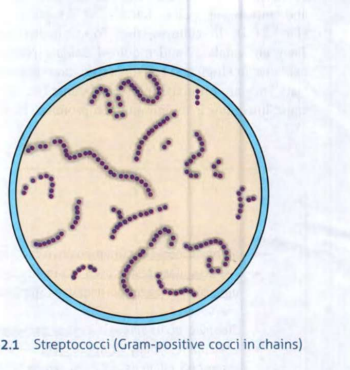


Fig. 22.1 Streptococci (Gram-positive cocci in chains)

Classification

Basis	Groups
Haemolysis on blood agar (Brown, 1919; 5% horse blood agar)	Alpha (α) : greenish discolouration, partial haemolysis, narrow zone (1–2 mm), intact RBCs visible — viridans streptococci, pneumococcus. Beta (β) : clear, colourless zone 2–4 mm, complete lysis — most pathogens. Gamma : non-haemolytic — mostly enterococci
Lancefield grouping (1933) — cell wall C carbohydrate	20 groups: A–H and K–V (no I, J)
Griffith typing — M protein of group A	~80 M types (M1, M2, M3...)
Biochemical reactions	Mainly to differentiate viridans streptococci
Oxygen requirement	Facultative anaerobes (<i>S. pyogenes</i>); obligate anaerobes (<i>Peptostreptococcus</i>)

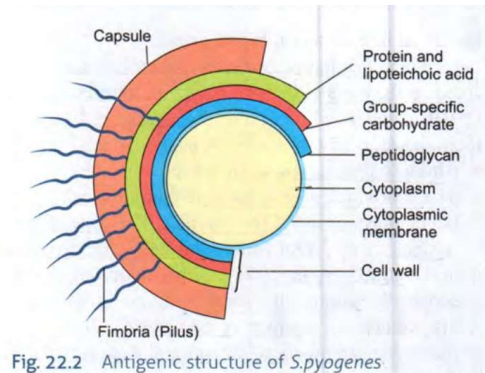
Species	Group	Haemolysis	Key lab test	Diseases
<i>S. pyogenes</i>	A	Beta	Bacitracin sensitive, PYR + , ribose not fermented	Pharyngitis, pyoderma, rheumatic fever, AGN
<i>S. agalactiae</i>	B	Beta	CAMP test, hippurate hydrolysis	Neonatal meningitis, septicaemia
<i>S. equisimilis</i>	C	Beta	Ferments ribose & trehalose	Pharyngitis, endocarditis
Enterococcus	D	Gamma (α, β)	Grows in 6.5% NaCl , PYR +	UTI, endocarditis
<i>S. bovis</i>	D	Gamma	No growth in 6.5% NaCl	Endocarditis
Viridans group	Not typed	Alpha	Optochin resistant	Endocarditis (<i>S. sanguis</i>), caries (<i>S. mutans</i>)

Adapted from Table 22.1 of the textbook.

2. STREPTOCOCCUS PYOGENES (GROUP A)

Antigenic Structure

- Cell wall: **outer layer** — **protein & lipoteichoic acid**; **middle** — **group-specific carbohydrate**; **inner** — **peptidoglycan** (rigidity; pyrogenic & thrombolytic).
- Capsule** (hyaluronic acid): inhibits phagocytosis; **not antigenic** in humans (resembles host hyaluronic acid).
- C carbohydrate antigen** → Lancefield group A. Must be extracted: **Lancefield's HCl method**, **Fuller's formamide**, **Maxted's enzyme** (*Streptomyces albus*), **Rantz & Randall autoclaving**; precipitation in capillary tubes within 5 min.
- M protein**: most important — **virulence (antiphagocytic)** and **typing**; antibody to M is **protective** and type-specific. Heat- and acid-stable, **trypsin-sensitive**. ~80 types.
- T protein**: acid-labile, **trypsin-resistant**; not related to virulence. **R protein** — no virulence role. **MAP** = M-associated protein.
- Pili (fimbriae)**: partly M protein, covered with **lipoteichoic acid** → attachment to epithelium (fibronectin receptor).



- Antigenic cross-reactions** → basis of sequelae: capsular hyaluronic acid ↔ synovial fluid; **cell wall protein** ↔ **myocardium**; **group A carbohydrate** ↔ **cardiac valves**; cytoplasmic membrane ↔ vascular intima; peptidoglycan ↔ skin.

Toxins and Enzymes

Factor	Key features
Streptolysin O	Oxygen-labile , heat-labile; activity only in subsurface colonies; antigenic → ASO ; cardiotoxic, leucotoxic; inhibited by cholesterol . Resembles O ₂ -labile lysins of <i>C. perfringens</i> , <i>C. tetani</i> , pneumococcus
Streptolysin S	Oxygen-stable , soluble in serum; causes surface haemolysis on blood agar; non-antigenic . (O and S also made by groups C and G)
Pyrogenic exotoxin (SPE)	Erythrogenic / Dick / scarlatinal toxin → scarlet fever . Types A, B, C (A & C phage-coded). Superantigens
Streptokinase (fibrinolysin)	Activates plasminogen → lyses fibrin barrier, spread of infection; antigenic. Used IV for myocardial infarction , thromboembolism
DNase (streptodornase)	Types A–D; B most antigenic → anti-DNase B (useful after skin infection). Liquefies thick pus; with streptokinase used to liquefy empyema
Hyaluronidase	"Spreading factor"; antigenic. Strains making large amounts (M 4, 22) are non-capsulated
NADase, Serum opacity factor (SOF)	NADase — leucotoxic?; SOF — lipoproteinase causing opacity of horse/swine serum agar

MEMORY AID

- Streptolysin **O** = **O**xygen-labile, **a**ntigenic (**ASO**). Streptolysin **S** = **S**table, **S**erum-soluble, **S**urface haemolysis, **S**ilent (non-antigenic).

Diseases Caused by *S. pyogenes*

Suppurative

- **Respiratory: sore throat** (commonest) — tonsillitis (older children, adults) or pharyngitis (younger children); complications — otitis media, mastoiditis, **quinsy**, **Ludwig's angina**, suppurative adenitis, rarely meningitis. Scarlet fever with SPE-producing strains.
- **Skin & soft tissue:** wound/burn infection, lymphangitis, cellulitis. **Erysipelas** — diffuse infection of superficial lymphatics; red, swollen skin **sharply demarcated**; older patients. **Impetigo** — young children; **higher-numbered M types**; with infected scabies → main cause of **AGN in tropics**.
- **Necrotising fasciitis** ("flesh-eating bacteria") — M types 1 & 3 with SPE-A; toxic shock-like syndrome, DIC. **Streptococcal TSS** (M 1, 3, 12, 28). **Puerperal sepsis** — Semmelweis (1847), hand washing. Abscesses, septicaemia, pyaemia.

Non-suppurative sequelae (1–4 weeks after infection)

Feature	Acute rheumatic fever	Post-streptococcal glomerulonephritis
Primary site	Throat	Throat or skin
Prior sensitisation	Essential	Not necessary
Serotype	Any	Nephritogenic: pyoderma 49, 53–55, 59–61; throat 1, 12
Immune response	Marked	Moderate
Complement level	Unaffected	Lowered
Repeated attacks	Common	Absent
Penicillin prophylaxis	Essential	Not indicated
Course / prognosis	Progressive or static / variable	Spontaneous resolution / good

- **Rheumatic fever:** carditis with **Aschoff nodules**; hypersensitivity / autoimmunity via cross-reacting antigens. **AGN:** antigenic cross-reaction with glomerular membrane or, more often, **immune complex disease**.

Epidemiology

- Colonises throat, nasopharynx, nose; carrier rate up to **20%**. Spread by direct contact, fingers, dust, fomites; skin infection spread by the eye gnat *Hippelates*. Throat infections commonest at **5–8 years**; crowding important. **Immunity is type-specific** (anti-M antibody). Typing by M/T/R antisera or **emm gene sequencing**.

Laboratory Diagnosis

- **Specimen:** throat swab, pus, exudate (culture immediately); **serum** for rheumatic fever & AGN. Transport: **Pike's medium** (blood agar + 1:1,000,000 crystal violet + 1:16,000 sodium azide).
- **Microscopy:** Gram-positive cocci in chains (useful for pus; **no value for throat/genital swabs**).
- **Culture:** **5% sheep blood agar**, 37 °C, anaerobically or 5–10% CO₂. Colonies small (0.5–1 mm), circular, semitransparent, with **wide zone of clear β-haemolysis**. Virulent = **matt**, avirulent = glossy, capsulated = mucoid. Liquid media: **granular turbidity with powdery deposit**. Selective: crystal violet + nalidixic acid + colistin blood agar. Storage: Robertson's cooked meat.



Fig. 22.3 Blood agar: showing beta hemolysis



Fig. 22.4 Blood agar: *S. pyogenes* magnified to show small colonies surrounded by zones of clear hemolysis

- **Identification:** non-motile, non-sporing; **catalase negative**; not bile soluble. **Bacitracin sensitivity** (0.04 U disc, Maxted) — wide zone with *S. pyogenes*. **PYR positive**, ribose not fermented. **Rapid group A antigen kits** from throat swabs (1–4 h; specific, less sensitive).

- **Serology (retrospective):** **ASO > 200 units** significant (latex agglutination now; high in rheumatic fever, often low in AGN/pyoderma). **Anti-DNase B > 300** and antihyaluronidase — best for **pyoderma**. **Streptozyme** — passive slide haemagglutination; positive after throat or skin infection.

Treatment & Prophylaxis

- **Penicillin G** — drug of choice (all group A sensitive); erythromycin or cephalexin in allergy. **Tetracyclines & sulphonamides not recommended**. Antibiotics have no effect on established AGN or rheumatic fever.
- **Long-term penicillin prophylaxis** in rheumatic fever to prevent re-infection and further heart damage; **not** needed for AGN (single episode, no re-infection).

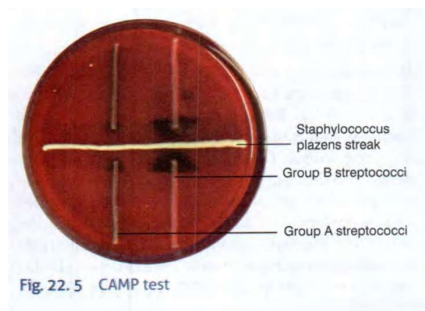
🔗 CLINICAL CORRELATION

- Sore throat → 2–3 weeks later joint pain + carditis = **rheumatic fever** (↑ ASO).
- Skin infection → 1–2 weeks later cola-coloured urine, oedema, hypertension = **AGN** (↑ anti-DNase B, ↓ complement).
- ASO test is of limited value in pyoderma — use **anti-DNase B**.

3. OTHER STREPTOCOCCI

Group B — *S. agalactiae*

- Causes **bovine mastitis**; in humans the **commonest cause of neonatal meningitis** (West). **Early onset** (within 1 week; meningitis/septicaemia; from maternal vagina) and **late onset** (2–12 weeks; from environment). Also sepsis, UTI in adults. 9 capsular serotypes.
- Identification: β-haemolysis; **hippurate hydrolysis**; **CAMP test** — accentuated haemolysis when streaked perpendicular to *S. aureus* (β-lysin).

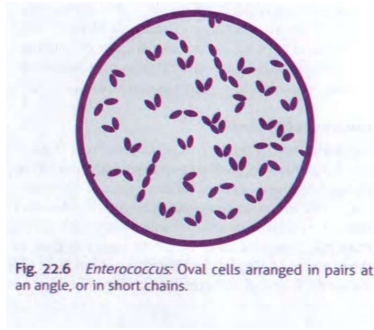


Groups C, F, G, D and others

- **Group C** (*S. equisimilis*): URTI, endocarditis, osteomyelitis; often **penicillin-tolerant** (add gentamicin); ferments ribose & trehalose; source of **therapeutic streptokinase**.
- **Group F**: "minute streptococci"; includes **Streptococcus MG** (primary atypical pneumonia test). **Group G**: tonsillitis, endocarditis, UTI.
- **Group D non-enterococcal** (*S. bovis*): grows on bile, hydrolyses **esculin**, **no growth in 6.5% NaCl**; UTI, endocarditis. Groups H, K (endocarditis), O, R (pigs).

Enterococcus

- Separate genus: *E. faecalis* (commonest), *E. faecium*, *E. durans*. Grow in **40% bile**, **6.5% NaCl**, **pH 9.6**, **45 °C**, **0.1% methylene blue milk**; survive **60 °C for 30 min**.
- **Oval cocci in pairs at an angle** or short chains; usually non-haemolytic; **tiny magenta-pink colonies on MacConkey**; *E. faecalis* gives black colonies on tellurite blood agar.
- UTI, wound infection, endocarditis, biliary infection, intra-abdominal abscess. **Intrinsically resistant to cephalosporins**; low-level aminoglycoside resistance → test **high-level gentamicin**; penicillin + aminoglycoside synergy. **VRE** (VanA–E; altered D-alanyl-D-alanine).



Viridans Group & NVS

- Alpha-haemolytic (greening) commensals of mouth and upper respiratory tract; **not Lancefield-groupable; optochin and bile resistant**. *S. mitis*, *S. mutans*, *S. salivarius*, *S. sanguis*.
- **Subacute bacterial endocarditis** (*S. sanguis* commonest) after dental extraction in damaged/prosthetic valves → **antibiotic prophylaxis** before dental procedures. *S. mutans* → **dental caries** (glycocalyx, acid from sugar).
- **Nutritionally variant streptococci (NVS)**: need **pyridoxal or cysteine**; culture-negative endocarditis, brain abscess.

★ GOLDEN ONE-LINERS

- Streptococci are **catalase negative** (staphylococci positive).
- *S. pyogenes*: **bacitracin sensitive, PYR positive**; *S. agalactiae*: **CAMP +**, hippurate +.
- M protein = virulence + typing; **T protein trypsin-resistant**.
- ASO > **200**; anti-DNase B > **300**.
- Streptolysin S causes surface β -haemolysis; streptolysin O is antigenic.
- Enterococci grow in **6.5% NaCl** and survive **60 °C for 30 min**.
- Commonest cause of SABE — ***S. sanguis***; dental caries — ***S. mutans***.

🎯 EXAM FOCUS

- Classification of streptococci (haemolysis + Lancefield) — always asked.
- Draw antigenic structure of *S. pyogenes* (Fig. 22.2).
- Rheumatic fever vs AGN table; laboratory diagnosis of rheumatic fever (ASO, anti-DNase B, streptozyme).
- Toxins of streptococci — streptolysin O vs S.

📄 EXAM QUESTIONS (from the textbook)

Essays

1. Classify streptococci and write a note on their pathogenesis.
2. Enumerate the virulence factors of streptococci and explain the laboratory diagnosis of streptococcal sore throat.
3. Draw a diagram of the cell wall of *S. pyogenes*; explain the relevance of cell wall antigens to virulence and classification.
4. Explain the pathogenesis and laboratory diagnosis of *S. pyogenes*.

Short answers / Short notes

1. Viridans streptococci and diseases caused by them; Streptococcus viridans
2. Classification of streptococci; diseases caused by streptococci; toxins of streptococci
3. Post-streptococcal (non-suppurative) complications
4. Laboratory diagnosis of rheumatic fever